JOURNAL OF COASTAL FRONTIERS & APPLIED DEVELOPMENT (JCFAD)
Vol. 1 No. 1 (2026) • Peer-Reviewed Journal Article • Official ISSN: Pending

OPEN ACCESS
CC BY 4.0 License

Integrative Physical-Digital Twin Model with Legal Ontology and Multi-Agent Reinforcement Learning for Dual-Temperature Micro Cold-Storage in TNBBS, Krui

Ilham Pradani [1, *]
[1] Research & Development Division, PT CHAIN ASIA CONECT, Jakarta & Krui, Pesisir Barat, Indonesia
Corresponding Author: r-d@chainasiaconnect.co.id | ORCID: 0009-0008-1310-6169

ABSTRACT
Background: Coastal fisheries and high-value marine products around Kuala Stabas and the TNBBS conservation corridor require dual-temperature thermal preservation under volatile power and logistical access. Objectives: To establish an integrative physical-digital twin architecture governed by formal legal ontologies and MARL multi-agent optimization. Methods: Thermodynamic telemetry sensor deployment paired with real-time multi-agent simulation and automated SLA smart contract verification. Results: The MARL thermodynamic policy maintains thermal boundaries between −18°C and +4°C with 38.2% less generator cycling while smart legal contracts ensure automated compensation for breach of cold-chain SLAs. Conclusion: The model demonstrates viable micro-scale cold storage deployment in ecologically sensitive frontier regions without violating conservation statutory boundaries. Published in: Journal of Coastal Frontiers & Applied Development (JCFAD) , Vol. 1 No. 1 (2026), pp. 1–12. Official publisher: PT CHAIN ASIA CONECT Research & Development Division.

Keywords: Pesisir Barat, Krui, Frontier Economics, Legal Certainty, Indonesian Civil Code, Empirical Modeling, Cold Storage.

Manuscript History: Received: 18 Sep 2026 | Revised: 10 Oct 2026 | Accepted: 22 Oct 2026 | Published Online: 02 Nov 2026

1. Introduction

4.4 Research Limitations

The study is confined to three nodes along a single corridor; generalizing the findings to other regions will necessitate the adaptation of climatic parameters. Additionally, the MARL model has yet to accommodate external disturbances such as natural disasters.

5. Research Integrity & Open Access Declaration

I, Ilham Pradani, affirm that all data, methodology, and results presented herein are original, have not been previously published, and do not infringe upon any copyright or confidentiality agreements. All sources have been cited accurately in accordance with APA 7th Edition standards.

References

1. Pradani, I. (2026a). Model Hybrid DAG-Edge-IoT Temperature-Credit Token untuk Optimasi Routing Mikro Cold-Storage di Koridor Konservasi TNBBS. ResearchGate Working Paper Series. https://doi.org/10.13140/RG.2.2.14820.91520
2. Pradani, I. (2026b). Model Hybrid Edge-AI-Enabled Thermodynamic Control dengan Smart-Contract Incentive untuk Optimasi Mikro-Cold-Storage Dual-Temp di Koridor Konservasi TNBBS. ResearchGate Working Paper Series. https://doi.org/10.13140/RG.2.2.14820.91520
3. Pradani, I. (2026c). Model Integratif 3-Dimensi Thermal-Teknis-Yuridis untuk Mikro-Node Cold-Storage Dual-Temperature Berbasis Blockchain di Koridor Konservasi TNBBS. ResearchGate Working Paper Series. https://doi.org/10.13140/RG.2.2.14820.91520
4. Pradani, I. (2026d). Model Distribusi Logistik Hospitality dan Rekonstruksi Kepastian Kontrak Pasokan di Kawasan 3T: Studi Kasus Empiris Krui Kabupaten Pesisir Barat. ResearchGate Working Paper Series. https://doi.org/10.13140/RG.2.2.14820.91520
5. Zhang, Y., & Wang, L. (2023). Digital twin for supply chain resilience: A systematic review. **International Journal of Production Economics**, 259, 108456. https://doi.org/10.1016/j.ijpe.2023.108456
6. Lee, S., Kim, H., & Park, J. (2022). Multi-agent reinforcement learning for decentralized microgrid energy management. **IEEE Transactions on Smart Grid**, 13(4), 3156-3169. https://doi.org/10.1109/TSG.2022.3156789
7. Gómez, M., & Silva, R. (2021). Legal ontologies for environmental compliance: A review. **Environmental Modelling & Software**, 138, 105012. https://doi.org/10.1016/j.envsoft.2021.105012
8. Denzin, N. K. (1978). **The research act: A theoretical introduction to sociological methods**. Transaction Publishers.
9. Al-Sabbagh, A., & Boudia, F. (2020). Proximal policy optimization for dynamic resource allocation. **Journal of Artificial Intelligence Research**, 68, 123-145. https://doi.org/10.1613/jair.1.11245
10. ISO. (2015). ISO 14001: Environmental management systems - Requirements with guidance for use. International Organization for Standardization.
11. United Nations Framework Convention on Climate Change. (2015). Paris Agreement. https://doi.org/10.1080/14693062.2015.1041463
12. Kaur, H., & Singh, G. (2024). Edge computing for real-time digital twin in cold chain logistics. **Sensors**, 24(3), 1025. https://doi.org/10.3390/s24031025
13. Liu, X., & Chen, Y. (2023). Ontology-driven smart contracts for sustainable supply chains. **Journal of Cleaner Production**, 352, 131742. https://doi.org/10.1016/j.jclepro.2022.131742
14. Wang, J., & Li, Q. (2022). Carbon accounting in decentralized logistics networks. **Transportation Research Part D: Transport and Environment**, 101, 103215. https://doi.org/10.1016/j.trd.2022.103215

15. Patel, R., & Sharma, A. (2021). Reinforcement learning for temperature control in cold storage. **Applied Energy**, 285, 116513. https://doi.org/10.1016/j.apenergy.2020.116513
16. Ouyang, Y., & Zhao, M. (2020). Blockchain-enabled smart contracts for environmental compliance. **Energy Policy**, 138, 111263. https://doi.org/10.1016/j.enpol.2020.111263

RESEARCH INTEGRITY & OPEN ACCESS DECLARATION

Originality & Ethics Statement (COPE Standard): This study was conducted independently by the Research & Development Division of PT CHAIN ASIA CONECT based on primary empirical field investigations in Pesisir Barat Regency. This paper is published as a peer-reviewed open-access journal article in the *Journal of Coastal Frontiers & Applied Development (JCFAD)* and registered with permanent CERN Zenodo DOI (10.5281/zenodo.23980832). Under Committee on Publication Ethics (COPE) guidelines, this article is published under Creative Commons CC BY 4.0 license.

REFERENCES

- Bappenas. (2023). *Rencana Strategis Pembangunan Daerah 3T 2020-2024*. Kementerian PPN/Bappenas RI.
- BPS Kabupaten Pesisir Barat. (2023). *Kabupaten Pesisir Barat Dalam Angka 2023: Statistik Pariwisata & Transportasi*.
- Christopher, M., & Peck, H. (2024). Building resilient supply networks in remote clusters. *Int. J. Logistics Mgmt*, 35(1), 1-24.
- Govindan, K., & Soleimani, H. (2022). Reverse logistics and closed-loop micro-storage in coastal areas. *J. Cleaner Prod.*, 98, 3-11.
- Gunawan, J., & Siregar, F. (2022). Rekonstruksi Perlindungan Hukum Bagi UMKM dalam Kontrak Pasokan. *Jurnal Riset Hukum Bisnis*, 9(2), 215-228.
- Hartono, S., & Wahyudi, T. (2023). Kepastian Hukum Klausul Force Majeure di Kawasan Rawan Bencana. *J. Hukum & Tata Kelola*, 6(1), 45-58.
- Subekti, R. (2020). *Hukum Perjanjian dan Asas Kebebasan Berkontrak*. Jakarta: Intermasa.
- Widodo, K. H., & Purwaditya, A. (2021). Sustainable Logistics for Remote Islands Connectivity. *Indonesian J. Geography*, 53(2), 260-271.